

Reinforcement Learning with Discriminative Process Supervision

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Abstract

Long-horizon agents often receive only an outcome reward after many tool calls, making it difficult to distinguish productive actions from the mistake that ultimately causes failure. We study whether a frozen, typed discriminative judge can supply useful transition-level measurements without training a task-specific process reward model. Our prefix-only protocol exposes public observations, actions, and static success criteria while withholding verifier outputs and oracle fields. Matched frozen-trace comparisons give heterogeneous evidence: Jev outperforms a local 8B language-model judge on ScienceWorld failure-step localization across three policy sizes, while naive score aggregation underperforms random best-of-five selection on two panels and ToolSandbox’s hidden milestones favor the language-model judge. We then test post-episode discriminative supervision in matched ALFWorld policy training. Jev sees the completed public trace and verified outcome and assigns each action the direct advantage $c(2q - 1)$, without adding the terminal GRPO advantage again. When all 16 initial trajectories fail, sparse GRPO gives zero advantage while Jev still produces nonconstant transition credit. After ten updates, held-out success is 21/64 for Jev versus 1/64 for sparse GRPO, with mean verifier scores 1.535 and 0.037. A historical outcome-anchored ablation reaches 22/64 but a lower mean score of 1.250. These one-seed results motivate replication but do not establish general sample efficiency or uniform superiority.

1. Introduction

Language-model post-training first made broad behavioral objectives learnable through reinforcement learning from

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human feedback (RLHF): human preferences supervise a learned reward model, and that reward model in turn supervises the policy (Ouyang et al., 2022). Reinforcement learning from AI feedback (RLAIF) subsequently replaced part of the human-labeling loop with model-generated judgments, improving the scalability of feedback while retaining a learned or generative evaluator (Lee et al., 2023). These approaches are flexible, but their flexibility comes from an evaluator that must itself be prompted, trained, or sampled.

Reinforcement learning with verifiable rewards (RLVR) offered a different path for domains such as mathematics and code, where an answer can be checked by an inexpensive external rule. DeepSeekMath introduced Group Relative Policy Optimization (GRPO), which estimates advantages from rewards of multiple responses to the same prompt without a learned critic (Shao et al., 2024); DeepSeek-R1 demonstrated how far this outcome-verification paradigm can scale reasoning (Guo et al., 2025). The resulting division of labor is compelling: the policy explores, and a verifier decides whether the result is correct. Yet a verifier normally answers only *whether* the completed response succeeded, not *which* decision caused that outcome.

The distinction becomes consequential as RL moves from single responses to interactive agents. Recent systems train policies over long sequences of reasoning, tool calls, observations, and environment changes (Chen et al., 2025; Luo et al., 2025). For K rollouts of one task, outcome-only GRPO uses an advantage of the form $A_i^{\text{out}} = (R_i - \bar{R})/(\sigma_R + \epsilon)$. If every rollout fails, then $R_1 = \dots = R_K$ and the entire group produces zero policy-gradient signal. If their outcomes differ, broadcasting A_i^{out} to every action in rollout i creates a second ambiguity: an early useful tool call and a later destructive action receive the same sign and magnitude. Thus long-horizon agentic RL has two coupled bottlenecks—*supervision availability* for outcome-equivalent failures and *temporal credit assignment* within each trajectory. Merely producing more labels is not sufficient: dense but constant or incorrectly localized rewards still provide poor credit.

Process supervision has developed in response to this gap. Human-labeled process reward models improve mathematical reasoning verification (Lightman et al., 2023), and agent-specific PRMs extend progress modeling to stateful interac-

tions (Choudhury, 2025; Xi et al., 2025). In parallel, agentic RL methods construct finer credit through state-conditioned grouping, information-gain potentials, reference-policy likelihoods, or hindsight reasoning (Feng et al., 2025; Wang et al., 2026; Tao et al., 2026; Xie et al., 2026; Tan et al., 2026). These advances establish the value of process credit, while exposing a practical design space: hand-designed verifiers are fast and trustworthy but usually terminal and task-specific; trained PRMs require suitable process data; and general LLM judges can interpret arbitrary traces but incur autoregressive inference and parsing overhead (Zheng et al., 2023). Recent evaluations further show that judging tool-using processes is a distinct problem from checking final answers (Li et al., 2026; Fan et al., 2026).

We introduce **RLDPS**, *Reinforcement Learning with Discriminative Process Supervision*, as a point between these regimes. RLDPS does not replace the outcome verifier. Instead, the verifier specifies and measures task success, while a frozen typed discriminative model decomposes the resulting experience into transition-level credit. After an episode, Jev receives the static success criteria, the completed public action–observation trace, and the verified outcome; for each transition it returns a continuous effect score $q_{i,t} \in [0, 1]$ and confidence $c_{i,t} \in [0, 1]$ through a typed numeric interface (TypeSafe AI, 2026a). We convert these judgments into advantages aligned only with the response tokens of the corresponding action. The annotator may use future observations and the terminal outcome because labeling occurs after collection and the policy never observes them; hidden oracle state, gold paths, and verifier internals remain excluded. In short, outcome verification determines *what happened*, whereas discriminative process supervision estimates *where credit belongs*.

This formulation treats density as a consequence of useful credit assignment rather than the objective itself. A failed trajectory can still contain beneficial transitions, and a successful trajectory can contain redundant or harmful ones. RLDPS therefore targets *credit recovery among outcome-equivalent trajectories*: it asks whether a lightweight general discriminator can separate those transitions well enough to create learning signal where outcome-only GRPO is silent, without paying the full generation cost of an LLM-as-a-judge pipeline. This is an empirical question rather than an architectural guarantee; Jev’s documented limitations on numerical precision, irrelevant context, and adversarial text require direct measurement (TypeSafe AI, 2026b).

Our contributions are threefold. First, we formulate and measure **failure-trajectory utilization**: on an initial ALF-World batch in which all 16 rollouts fail, binary outcome GRPO yields zero relative advantage, whereas discriminative process supervision assigns nonconstant transition-specific advantages. Second, we implement **transition-**

resolved credit assignment: continuous effect and confidence estimates become action-token advantages rather than a single trajectory score, and are evaluated by progress discrimination, first-fault localization, and downstream policy learning. In the current one-seed ten-update pilot, post-episode Jev reaches 21/64 held-out successes and mean verifier score 1.535, compared with 1/64 and 0.037 for sparse GRPO. Third, we study a **quality–efficiency trade-off** between programmatic RLVR and an autoregressive 8B LLM judge on 600 frozen trajectories and 5,743 transitions. The frozen results are deliberately mixed—Jev improves failure-step localization on ScienceWorld but is not uniformly superior in trajectory selection or hidden-milestone agreement—so our current evidence supports discriminative supervision as a promising source of process credit, not a universal judge. All policy results use one seed; claims of general sample efficiency remain contingent on replication.

2. Related Work

Outcome-based agentic reinforcement learning and delayed credit. GRPO replaces a learned value baseline with within-group reward comparisons (Shao et al., 2024). Its simple outcome formulation is useful when a verifier is trustworthy, but equal rewards produce no group-relative contrast. TRACE studies the separate long-horizon problem of assigning turn-level credit from a frozen reference model and reports training gains in search agents (Tao et al., 2026). Our judge-based question differs: we ask whether a *general-purpose frozen discriminator*, without a trained task-specific credit estimator, can retrospectively decompose a verified agent outcome into useful transition credit. We preserve continuous terminal rewards as an outcome-only control so that any benefit cannot be attributed merely to replacing a success bit with a more graded final score.

Process reward models and agent evaluation. Process supervision was shown to improve verification on mathematical solutions with human step annotations (Lightman et al., 2023). For agents, AgentPRM and related methods learn progress-sensitive signals from interaction or demonstrations (Choudhury, 2025; Xi et al., 2025). ToolPRMBench emphasizes local error discrimination in realistic tool-use histories and shows why intermediate judgment needs its own evaluation rather than being inferred from final-answer accuracy (Li et al., 2026). Our setting differs in using Jev as a zero-shot annotator: no PRM parameters are trained before scoring the frozen traces. This removes a PRM-training stage but does not remove the need to validate labels, guard against reward hacking, or test generalization during policy optimization.

Discriminative judging and measurable process credit. Jev accepts a state and independently evaluates typed ques-

tions, returning a distribution over descriptive Score levels or a Noul probability (TypeSafe AI, 2026a). We use this interface for a narrowly specified transition-effect judgment and retain the full response distribution. The vendor documents limitations on numerical precision, irrelevant context, and adversarial text (TypeSafe AI, 2026b); consequently, arithmetic is done outside the model and inputs are filtered to task-relevant public fields. Unlike a generic “rate this trajectory” prompt, this yields a falsifiable local signal whose agreement with hidden verifier anchors, temporal localization, latency, and downstream GRPO effects can be measured separately.

3. Methods

The frozen measurement study uses a prefix-only whitelist of the user task, explicit static success criteria, at most three earlier public action–observation pairs, the current public observation, action, and observed response. It excludes the trajectory’s verifier judgment, process rewards, oracle diagnostics, gold action sequences, and future steps. All offline judges score the same previously sampled trajectories, so no second policy rollout is generated. The training method asks a different, retrospective question after termination: Jev receives the success criteria, complete public trajectory, and verified terminal reward and success bit, with one typed Score question per transition. Oracle state, gold paths, stored process labels, and hidden verifier internals remain excluded. Both protocols use Jev’s native continuous expected Score $q_{i,t} \in [0, 1]$ and confidence $c_{i,t} \in [0, 1]$ without discretization.

The offline primary slice comprises groups whose five binary outcome rewards are all zero. We compare step labels against held-out native process anchors, report failure-only AUROC and first-fault error, and measure best-of-five terminal verifier utility without exposing those anchors to the prefix judge. Quality must be reported jointly with throughput and token cost. For training, sparse RLVR uses $R_i = 10 \mathbf{1}[\text{success}_i]$, standardizes the four trajectory returns for a task, and broadcasts the resulting A_i^{out} to every action in trajectory i . Our Jev arm assigns the direct turn-level advantage

$$A_{i,t} = c_{i,t}(2q_{i,t} - 1). \quad (1)$$

This advantage is not group-standardized: its magnitude carries confidence, and standardization would amplify nearly neutral judgments. The outcome is not added again as A_i^{out} because it already conditions the retrospective judgment. Only action tokens emitted at turn t receive $A_{i,t}$. With token-level importance ratio $\rho_{i,t,k}$, the actor minimizes the usual clipped PPO surrogate $-\min(\rho A_{i,t}, \text{clip}(\rho, 1 - \epsilon, 1 + \epsilon)A_{i,t})$. The policy never consumes future states, the judge, or the outcome at deployment. Held-out evaluation uses only the environment verifier. All controlled runs

start from Qwen2.5-1.5B-Instruct with seed 0, four prompts and four rollouts per update, a 30-action horizon, ten updates, and `valid_seen` evaluations at updates 0, 5, and 10. Prefix-only, outcome-anchored, same-turn group-relative, and trajectory-aggregate variants are retained only as historical ablations. These one-seed pilots are not a general sample-efficiency test.

4. Experiments

Table 1 summarizes the matched frozen-trace experiments before the detailed results. AUROC is computed within a task group or game and then averaged so that long trajectories do not dominate. The number of eligible units is reported because several panels contain many all-failure groups but few groups with both positive and negative native process events.

The first complete judge panel covers all 215 transitions of 100 frozen Qwen3-8B ToolSandbox rollouts. In its 16 all-failed groups, every one of the 80 trajectories receives a nonzero Jev-based trajectory RLOO advantage, whereas binary outcome GRPO receives zero. Yet density alone fails the quality check: on nonterminal transitions, the mean within-trajectory AUROC for detecting positive official milestone deltas is 0.304 for Jev across only 14 eligible traces (three task groups); the corresponding within-group AUROC is 0.353. A matched local, non-thinking Qwen3-8B language-model judge, given the same state and rubric, obtains 0.679 and 0.720, respectively. Observed mean latency is 0.908 seconds per remote Jev call versus 1.195 seconds per local LLM call; this is not a hardware-matched speed comparison. These small, correlated strata do not support a general performance claim. The mismatch is informative: some hidden ToolSandbox milestones give credit to an action whose public tool response reports an error or no visible state change. A judge restricted to public prefixes cannot observe that hidden update. We keep this negative result rather than revise the rubric on the same test groups.

The complete paired pass over 2,069 previously sampled Qwen3-8B ScienceWorld transitions shows a different pattern. Among 17 all-five-failed groups, mean AUROC for positive native score changes is 0.900 for Jev within 45 eligible trajectories versus 0.490 for the matched non-thinking Qwen3-8B language-model judge. Within ten eligible task groups, the means are 0.876 and 0.502, respectively; Jev is higher in nine groups. The paired group-mean difference is 0.374, with an exploratory task-bootstrap 95% interval of [0.236, 0.507] from 20,000 resamples. Both judges received identical public transition states and semantic criteria. An exploratory slice excluding repeated state–action pairs retains a mean group AUROC of 0.876 versus 0.416 over 627 steps, so the gap is not solely loop detection. Binary and continuous terminal rewards cannot localize different steps

Table 1. Summary of frozen-trace process-credit measurements. Each Jev/LLM pair scores the same public states and semantic criterion. “Groups” denotes eligible task groups or games for the reported equal-weighted AUROC.

Panel	Process target	Steps	Groups	Jev	Local 8B LLM
ToolSandbox, Qwen3-8B	positive official milestone delta	215	3	0.353	0.720
ScienceWorld, Qwen3-8B	positive native score delta	2,069	10	0.876	0.502
ScienceWorld, Qwen3-4B	positive native score delta	1,578	7	0.865	0.560
ScienceWorld, Qwen3-1.7B	positive native score delta	1,596	3	0.989	0.388
ALFWorld dev, Qwen3-8B	shortest-plan progress	306	5	0.892	0.797
ALFWorld dev, Qwen3-8B	hard-step quality	306	6	0.773	0.660
ALFWorld failures, Qwen3-1.7B	shortest-plan progress	300	4	0.563	0.628
ALFWorld failures, Qwen3-1.7B	hard-step quality	300	5	0.761	0.551

inside a fixed trajectory (AUROC 0.500 by construction), although continuous final reward remains a cross-trajectory control. Jev yields nonzero trajectory-level RLOO advantage on 83 of the 85 failed-group trajectories and varying reward-to-go on 64. This establishes a process-measurement advantage over this particular 8B judge on the eligible ScienceWorld tasks, not over all judges or in downstream learning.

A second complete pass on 1,578 frozen Qwen3-4B ScienceWorld transitions replicates the local-credit direction across policy sizes, with the same judge and rubric. In 16 all-five-failed groups, seven task groups have discriminable native score-change events; their mean within-group AUROC is 0.865 for Jev versus 0.560 for the LLM, with Jev higher in six groups. The paired mean difference is 0.305 and an exploratory task-bootstrap 95% interval is [0.126, 0.493]. Across 25 eligible traces, mean within-trace AUROC is 0.809 versus 0.482. Jev has higher nonzero step-wise RLOO density among failed trajectories (0.988 versus 0.579), but density is not substituted for the AUROC result. Mean observed remote Jev latency is 1.015 s versus 0.848 s for the local LLM; hardware and network are unmatched. The policy panels share benchmark tasks, so they are not independent task replications.

A third complete pass covers 1,596 frozen Qwen3-1.7B ScienceWorld transitions. Nineteen of 20 five-rollout groups all fail, so binary outcome GRPO assigns zero group advantage to all 95 trajectories in that slice. Only three groups contain both positive and non-positive transitions. Mean within-group AUROC for detecting positive native score changes on those three groups is 0.989 for Jev versus 0.388 for the LLM; the paired difference is 0.601 with an exploratory task-bootstrap 95% interval of [0.427, 0.867]. Jev wins all three groups. Across 15 eligible traces the corresponding means are also 0.989 and 0.388. Failure-only nonzero step-wise RLOO density is 0.991 versus 0.346. An exploratory control removes repeated state-action pairs, leaving 298 failed-group steps; Jev remains higher at 0.882 versus 0.262, so loop detection alone does not explain the gap. The near-ceiling primary result is strong local evidence but has narrow task support. Moreover, all sampled failed rollouts within

each group have equal continuous verifier reward, so this panel cannot test best-of-five ranking utility.

The predeclared best-of-five test sharply limits the reward claim. On the 17 all-failed 8B-policy groups, selecting a rollout by the *sum* of Jev transition scores yields mean official continuous final reward 0.079, below random selection (0.120), the matched LLM judge (0.129), and the sampled oracle maximum (0.151). Averaging Jev scores also yields 0.079. The 4B-policy panel repeats this negative trajectory-level result: Jev selects 0.010 under either sum or mean aggregation, below random 0.0196 and sampled oracle 0.0394. Thus local progress discrimination does not justify naive additive trajectory reward; the signed Jev signal may penalize long unsuccessful exploration more than the official partial-credit verifier does. Any training reward transformation must be chosen on development tasks and assessed on held-out tasks, not tuned to reverse this observed test result.

An exploratory ALFWorld development slice confirms the aggregation issue without establishing a training gain. Complete Jev annotations of 306 transitions from 12 frozen Qwen3-8B traces across six `valid_unseen` games yield mean within-game AUROC 0.892 for shortest-plan progress (five eligible games) and 0.773 for hard-step quality (six games); a matched local non-thinking Qwen3-8B judge scores the exact same public states and obtains 0.797 and 0.660. A paired game-bootstrap gives exploratory 95% intervals of [-0.007, 0.220] and [0.031, 0.209] for the two AUROC differences, respectively. First-hard-error exact localization is 6/12 for Jev versus 1/12 for this LLM. Mean observed call latency is 1.136 s for remote Jev and 0.871 s for the local LLM, which does not establish a hardware-matched speed advantage. In best-of-two selection, the mean centered Jev and LLM scores each pick a successful trace in three of six games, the sampled oracle ceiling, whereas the sum of Jev scores picks one of six and random selection has expectation two of six. The slice is too small for a general claim and cannot be reused as independent downstream test data after informing the reward aggregation choice.

An earlier, independent ALFWorld hard-failure set provides a contrasting diagnostic. On the same 300 frozen transitions from ten failed Qwen3-1.7B trajectories across five games, pooled failure-only shortest-plan-progress AUROC is 0.500 for terminal RLVR, 0.650 for Jev, and 0.629 for the matched LLM judge. Yet equal-weighted within-game AUROC is 0.500, 0.563, and 0.628 across four eligible games: the apparent Jev advantage over the LLM does not survive task balancing. For hard-step quality, equal-weighted within-game AUROC is 0.500, 0.761, and 0.551 across all five games, with Jev above the LLM in each game. Jev identifies the first hard error exactly in three of ten traces, versus zero for both comparators. The set is too small and correlated for a general superiority claim; these are offline measurement results, not evidence of improved GRPO learning.

The earlier trajectory-aggregate ALFWorld ablation starts both Qwen2.5-1.5B-Instruct arms at 0/64 held-out valid_{seen} success. All 16 trajectories fail in each of the first five training groups. After removing reward fields and random identifiers, the arms’ first 480 transition records have the same SHA-256 digest. The binary-outcome arm consequently has zero reward, zero advantage, and policy-gradient loss 0.000 at every update through five. The aggregate Jev arm yields nonzero trajectory advantages; at update 5, held-out success is 2/64 versus 1/64, and mean continuous verifier score is 0.101 versus 0.037. This one-success difference is inconclusive and the ablation does not assign distinct step advantages.

The current Jev-only run begins from 0/64 held-out success. Its first batch contains 480 public transitions and all 16 trajectories fail. Sparse GRPO therefore assigns zero advantage to every token and reports policy-gradient loss 0.000. Jev assigns nonzero advantage to 479/480 transitions, and all 16 trajectories have nonconstant within-trajectory advantage. This establishes that distinct $A_{i,t}$ values reach the PPO objective rather than merely appearing in an annotation log. Across all ten updates, 4,363 transitions are labeled, 99.51% receive nonzero advantage, and the nonconstant-trajectory fraction remains 1.000. Sparse GRPO receives an active outcome signal in only one of ten updates.

The current Jev-only run reaches 16/64 success at update 5 and 21/64 at update 10, compared with 0/64 and 1/64 for sparse GRPO (Table 2). Its final mean continuous verifier score is 1.535 versus 0.037. This satisfies the predeclared pilot gates—supervision under all-failure outcomes, dense turn-specific advantages that enter PPO, and a better trained-policy benchmark result. The magnitude justifies a formal multi-seed study, but one seed and 64 held-out tasks cannot establish a general Jev advantage, variance across training seeds, or sample-efficiency scaling.

The historical prefix-only run reaches 19/64 without seeing future steps or the outcome, and therefore remains the

causal-prefix measurement setting. The historical outcome-conditioned run adds $0.1A_i^{\text{out}}$ to the same Jev credit and reaches 22/64, one success above Jev-only, but its final continuous score is lower (1.250 versus 1.535). Since the outcome anchor is zero when all four rollouts fail, the anchored and Jev-only variants share the same first-batch 479/480 nonzero transition credit. Their endpoint ordering is mixed and too small to identify a reliable advantage from adding the terminal signal again.

The current run’s on-policy training-batch success counts are [0, 0, 4, 1, 6, 1, 2, 0, 4, 9] out of 16. These counts are descriptive and not held-out evaluation. One completed-trajectory request contains all step questions, reducing HTTP request count relative to the historical one-request-per-transition prefix implementation; each request is larger, so call count alone is not a controlled throughput or cost comparison.

The superseded same-turn group-relative implementation was audited separately. It produced dense turn-level advantages in an all-failure batch but reached only 2/64 success and mean score 0.106 at update 10. The earliest trajectory-aggregate variant is also retained only as an ablation because it broadcasts one scalar to every action and therefore does not implement within-trajectory credit.

5. Discussion and limitations

What the experiments establish. The most direct result is mechanical but important: when all 16 sampled trajectories fail, sparse binary GRPO has no within-group variance, while Jev assigns nonconstant turn-level advantages that reach the clipped PPO loss. This resolves the signal-availability part of the delayed-credit problem. The frozen ScienceWorld panels additionally show that the prefix-only signal can align with native positive score changes substantially better than one matched local 8B judge across three policy sizes. Finally, the current post-episode Jev-only arm outperforms sparse GRPO on the same 64 held-out ALFWorld tasks after both five and ten updates. These claims concern the tested model, rubric, tasks, and seed.

What the experiments do not establish. Density is not accuracy. ToolSandbox reverses the ScienceWorld judge ordering because its hidden milestones are not always inferable from public observations, and naive sums or means of Jev scores underperform random best-of-five selection on two ScienceWorld panels. The historical outcome-anchored arm has one more strict success than Jev-only but a lower continuous verifier score, while same-turn centering fails badly. The available evidence therefore rejects a universal “Jev is a better reward” claim. A more precise claim is that Jev can provide useful local credit when the success predicate and observable transition expose the relevant state

Table 2. Matched held-out ALFWorld `valid_seen` evaluation. All arms start from the same checkpoint, use seed 0, and evaluate the same 64 tasks. Historical variants remain frozen.

Evaluation	Sparse	Prefix	Outcome+anchor	Jev-only
Before training, success	0/64	0/64	0/64	0/64
Update 5, success	0/64	8/64	16/64	16/64
Update 5, mean score	0.000	0.427	1.214	1.067
Update 10, success	1/64	19/64	22/64	21/64
Update 10, mean score	0.037	1.423	1.250	1.535

change; the mapping from local judgments to an optimization reward remains task- and conditioning-dependent.

Causal prefix credit versus hindsight labels. The prefix protocol tests whether a transition can be judged using only information available at that point. The training method instead asks a different and often easier question: knowing the complete public trajectory and verified outcome, which earlier transitions deserve positive or negative policy gradient? Using future information in a post-episode label is not a policy inference mismatch—RL returns and retrospective credit estimators do the same—but it prevents claims about real-time scoring or causal prefix quality. We therefore retain prefix-only frozen benchmarks as the causal measurement test and use outcome-conditioned labels only for post-episode training.

Statistical and systems limits. All policy-training results use one seed, ten updates, one small policy checkpoint, and 64 held-out `valid_seen` tasks. The three ScienceWorld policy-size panels reuse benchmark tasks and are not independent task replications. The Jev service is remote and closed, while the comparison LLM is local; observed latency is neither hardware-matched nor a normalized dollar comparison. The hindsight request reduction counts HTTP calls rather than equal-sized units of work. Formal claims about sample efficiency, cost, or general benchmark superiority require multiple training seeds, broader task families, paired per-task uncertainty, and controlled serving measurements.

6. Conclusion

Across frozen traces and policy training, Jev is most promising as a source of dense signed transition credit when sparse outcome groups collapse to a common reward. Prefix-only labels recover meaningful native process events on ScienceWorld, while the maintained post-episode Jev-only method retains dense optimizer signal and produces a large one-seed ALFWorld gain over sparse GRPO. The evidence is deliberately mixed: hidden-state ToolSandbox labels and trajectory-level selection expose failure modes, and historical reward transformations do not improve every held-out metric. The next decisive experiment is multi-seed replication of sparse GRPO and the single maintained Jev method,

not further post-hoc rubric tuning on the current tasks.

References

- Chen, K., Cusumano-Towner, M., Huval, B., Petrenko, A., Hamburger, J., Koltun, V., and Krähenbühl, P. Reinforcement learning for long-horizon interactive llm agents. *arXiv preprint arXiv:2502.01600*, 2025.
- Choudhury, S. Process reward models for LLM agents: Practical framework and directions, 2025. URL <https://arxiv.org/abs/2502.10325>.
- Fan, S., Ye, X., Huo, Y., Chen, Z.-Y., Guo, Y., Yang, S., Yang, W., Ye, S., Chen, J., Chen, H., Cong, X., and Lin, Y. AgentProcessBench: Diagnosing step-level process quality in tool-using agents, 2026. URL <https://arxiv.org/abs/2603.14465>.
- Feng, L., Xue, Z., Liu, T., and An, B. Group-in-group policy optimization for LLM agent training. *arXiv preprint arXiv:2505.10978*, 2025.
- Guo, D., Yang, D., Zhang, H., Song, J., Zhang, R., Xu, R., Zhu, Q., Ma, S., Wang, P., Bi, X., et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Lee, H., Phatale, S., Mansoor, H., Mesnard, T., Ferret, J., Lu, K., Bishop, C., Hall, E., Carbune, V., Rastogi, A., et al. Rlaif: Scaling reinforcement learning from human feedback with ai feedback. *arXiv preprint arXiv:2309.00267*, 2023.
- Li, D., Yao, Y., Tan, Z., Liu, H., and Guo, R. ToolPRM-Bench: Evaluating and advancing process reward models for tool-using agents, 2026. URL <https://arxiv.org/abs/2601.12294>.
- Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., Leike, J., Schulman, J., Sutskever, I., and Cobbe, K. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.
- Luo, X., Zhang, Y., He, Z., Wang, Z., Zhao, S., Li, D., Qiu, L. K., and Yang, Y. Agent Lightning: Train any AI agents with reinforcement learning, 2025. URL <https://arxiv.org/abs/2508.03680>.

- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y., Wu, Y., et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Tan, H.-Z., Yang, X.-W., Chen, H., Shao, J.-J., Wen, Y., Shen, Y., Luo, W., Du, X., Guo, L.-Z., and Li, Y.-F. Hind-sight credit assignment for long-horizon LLM agents. *arXiv preprint arXiv:2603.08754*, 2026.
- Tao, L., Peng, B., Yao, W., et al. TRACE: Turn-level reward assignment via credit estimation for long-horizon agents, 2026. URL <https://arxiv.org/abs/2607.13988>.
- TypeSafe AI. Typesafe AI: Introduction and primitives documentation, 2026a. URL <https://docs.typesafe.ai/introduction>. Accessed 2026-09-19.
- TypeSafe AI. Jev 1.13 jaggedness, 2026b. URL <https://docs.typesafe.ai/model-jaggedness/jev-1.13>. Accessed 2026-09-19.
- Wang, G., Dai, S., Ye, G., Gan, Z., Yao, W., Deng, Y., Wu, X., and Ying, Z. Information gain-based policy optimization: A simple and effective approach for multi-turn search agents. In *International Conference on Learning Representations*, 2026.
- Xi, Z., Liao, C., Li, G., et al. AgentPRM: Process reward models for LLM agents via step-wise promise and progress, 2025. URL <https://arxiv.org/abs/2511.08325>.
- Xie, Y., Thomas, N., Hansen, N., Fu, Y., Li, L. E., and Wang, X. TIPS: Turn-level information-potential reward shaping for search-augmented LLMs, 2026. URL <https://arxiv.org/abs/2603.22293>.
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems*, 36:46595–46623, 2023.